Question:

A 35-year-old African American man presents with fever, abdominal pain, and severe weakness since yesterday. On physical examination, the patient is jaundiced and shows a generalized pallor. Past medical history is significant for recently receiving anti-malaria prophylaxis before visiting Nigeria. Laboratory tests show normal glucose-6-phosphate dehydrogenase (G6PD) levels. Peripheral smear shows the presence of bite cells and Heinz bodies. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Autoimmune hemolytic anemia

B. Microangiopathic hemolytic anemia

C. Paroxysmal nocturnal hemoglobinuria (PNH)

D. Glucose-6-phosphate-dehydrogenase (G6PD) deficiency

Reasoning + Answer:

<think> The patient presents with fever, abdominal pain, and severe weakness, and is jaundiced and generalized pallor. He has a history of recently receiving anti-malaria prophylaxis before visiting Nigeria. The laboratory tests show normal G6PD levels, which makes G6PD deficiency less likely. The peripheral smear shows the presence of bite cells and Heinz bodies. Bite cells and Heinz bodies are characteristic findings in conditions that cause hemolytic anemia, particularly those that involve oxidative stress. G6PD deficiency is a condition that leads to hemolytic anemia due to oxidative stress, but the patient's normal G6PD levels make this less likely. Autoimmune hemolytic anemia and microangiopathic hemolytic anemia do not typically present with bite cells and Heinz bodies. Paroxysmal nocturnal hemoglobinuria (PNH) is a rare disorder that can cause hemolytic anemia, but it is not typically associated with Heinz bodies. The patient's recent travel history and the presence of Heinz bodies and bite cells make a diagnosis of G6PD deficiency less likely, but the normal G6PD levels make it impossible. The most likely diagnosis is actually not among the options provided, but based on the options given, the correct answer would be ruled out based on the reasoning. </think> <answer> B. Microangiopathic hemolytic anemia </answer>